SCSI 1013 DISCRETE STRUCTURE



CHAPTER 4 – PART 2

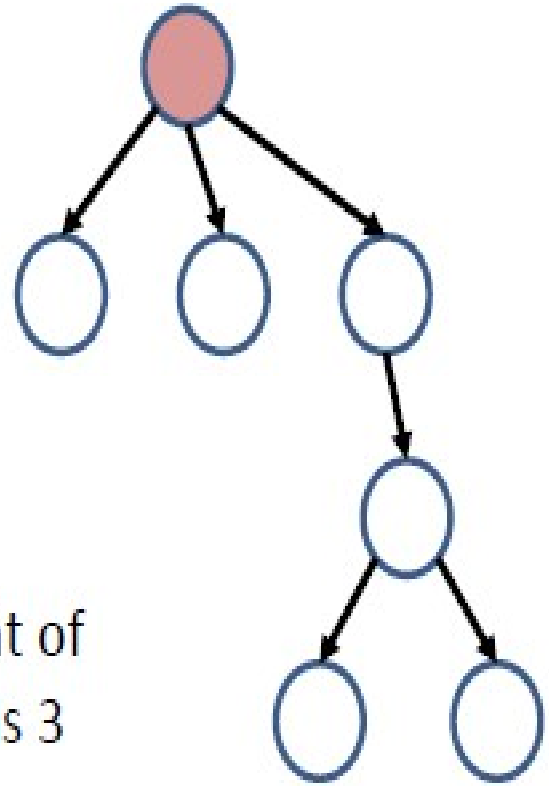
TREE

Semester 1 – 2017/2018

Properties of Trees

- The level of a vertex v in a rooted tree is the length of the unique path from the root to this vertex.
- The level of the root is defined to be zero.
- The height of a rooted tree is the maximum of the levels of vertices.

Example



Level 0

Level 1

Level 2

Level 3

This height of the tree is 3

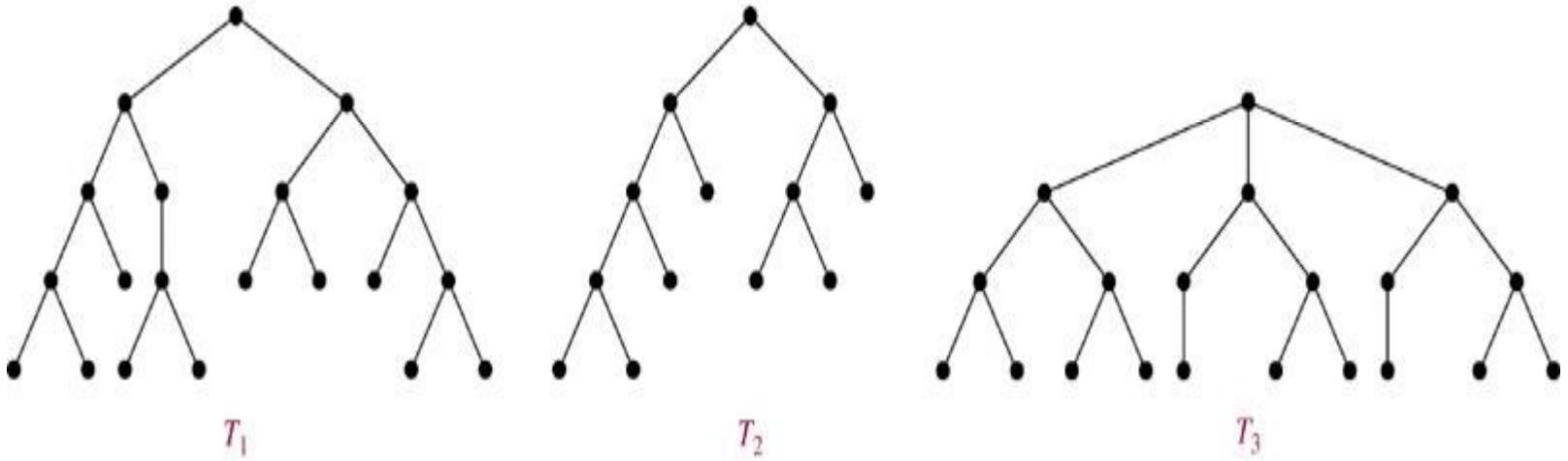
Properties of Trees

- **Definition:** A rooted m-ary tree of height h is balanced if all leaves are at levels h or $h-1$.

- **Theorem.** There are at most m^h leaves in an m-ary tree of height h.

Example

Which of the rooted trees shown below are balanced?



Sol. T_1, T_3

Tree Traversal

Universal Address Systems

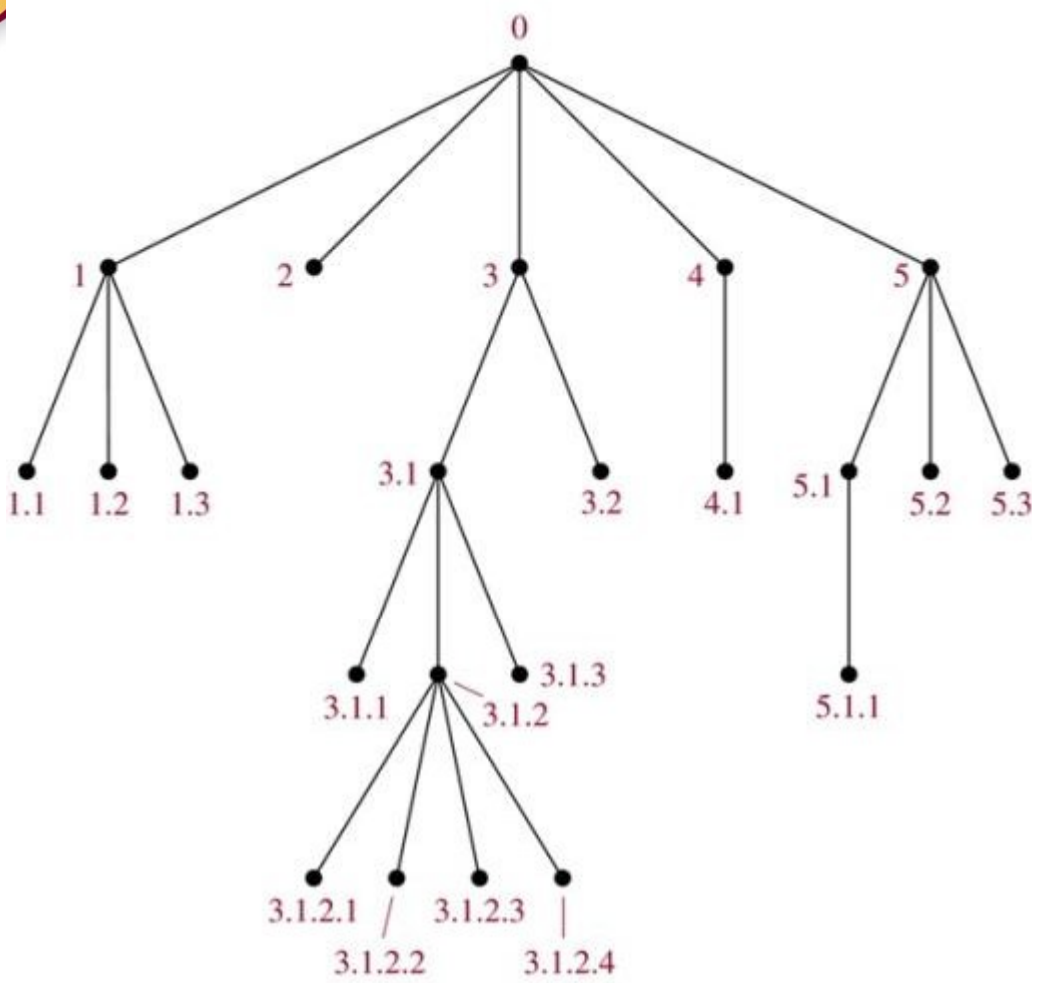
Label vertices:

1.root $\rightarrow$ 0, its k children $\rightarrow$ $1, 2, \ldots, k$ (from left to right)

2.For each vertex v at level n with label A,
its r children $\rightarrow$ $A.1, A.2, \ldots, A.r$ (from left to right).

We can totally order the vertices using the lexicographic ordering of their labels in the universal address system.

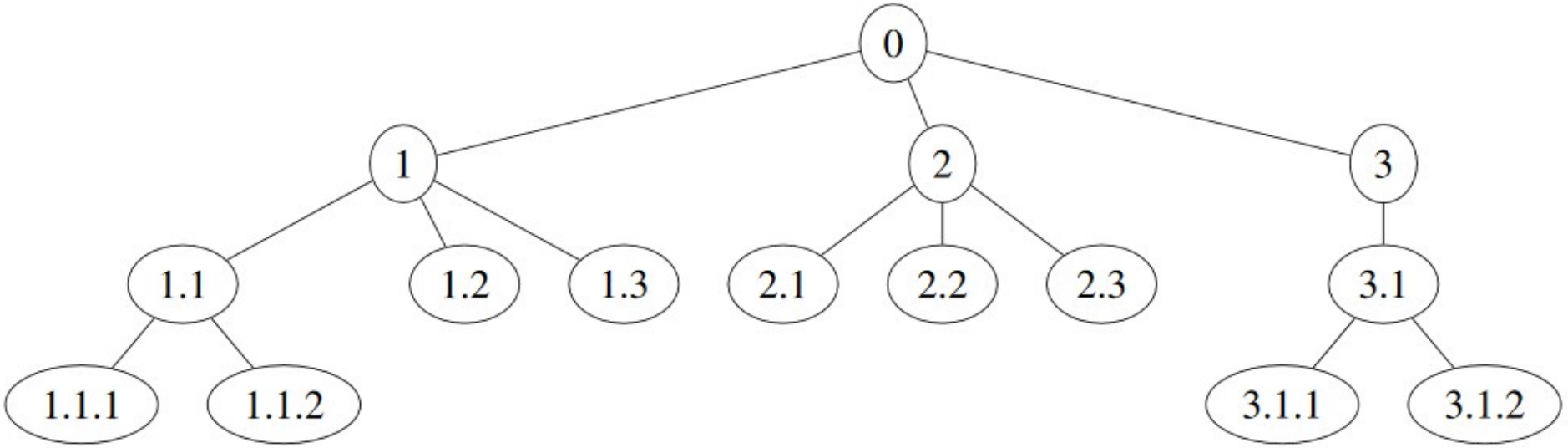
$x_1.x_2.\ldots.x_n < y_1.y_2.\ldots.y_m$
if there is an i, $0 \le i \le n$, with $x_1 = y_1$, $x_2 = y_2$, …, $x_{i-1} = y_{i-1}$, and $x_i < y_i$; or if $n < m$ and $x_i = y_i$ for $i = 1, 2, \ldots, n$.



The lexicographic ordering is:

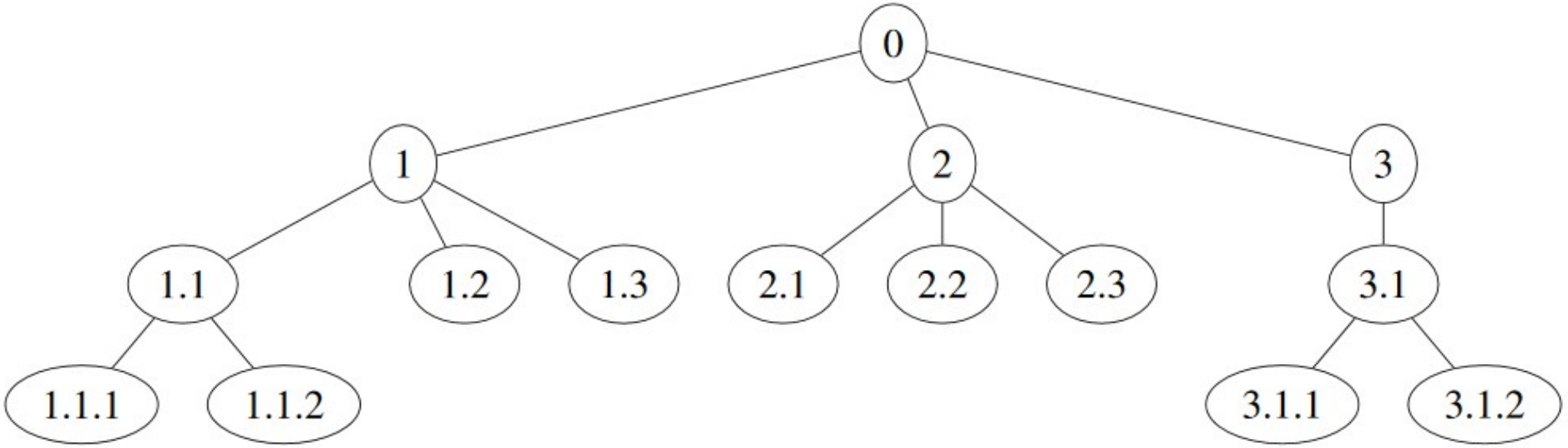
$0 < 1 < 1.1 < 1.2 < 1.3 < 2 < 3 < 3.1 < 3.1.1 < 3.1.2 < 3.1.2.1 < 3.1.2.2 < 3.1.2.3 < 3.1.2.4 < 3.1.3 < 3.2 < 4 < 4.1 < 5 < 5.1 < 5.1.1 < 5.2 < 5.3$

Exercise



Find the lexicographic ordering of the above tree.

Solution



The lexicographic ordering of the above tree:

$$0 < 1 < 1.1 < 1.1.1 < 1.1.2 < 1.2 < 1.3 < 2 < 2.1 < 2.2 < 2.3 < 3 < 3.1 < 3.1.1 < 3.1.2$$

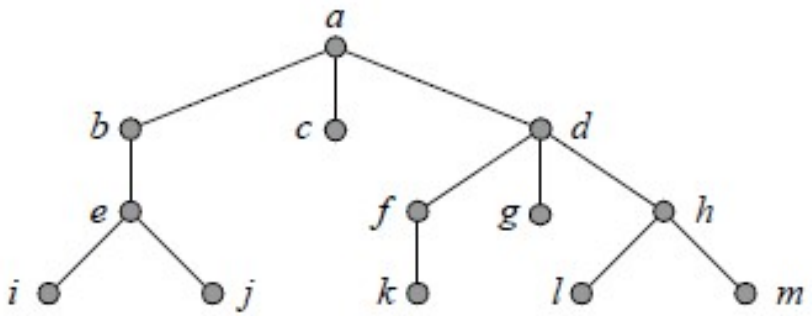
Tree Traversal

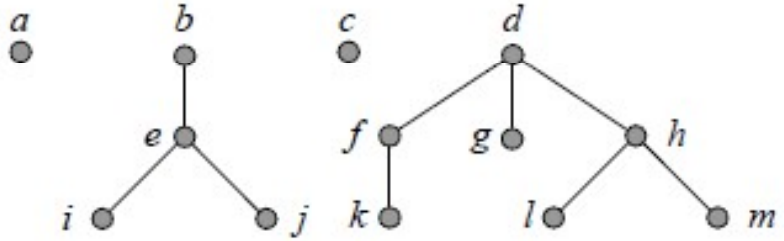
- Preorder: root, left-subtree, right subtree
- Inorder – left subtree, root, right sub-tree
- Post-order : left subtree, right sub-tree, root

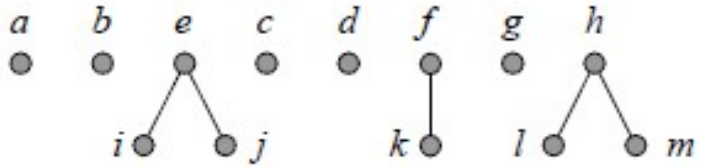
Preorder Traversal

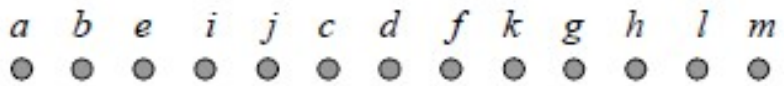
Procedure *preorder*(*T*: ordered rooted tree)
r := root of *T*
list *r*
for each child *c* of *r* from left to right
begin
    T(*c*) := subtree with *c* as its root
    preorder(*T*(*c*))
end

Preorder Traversal - Example

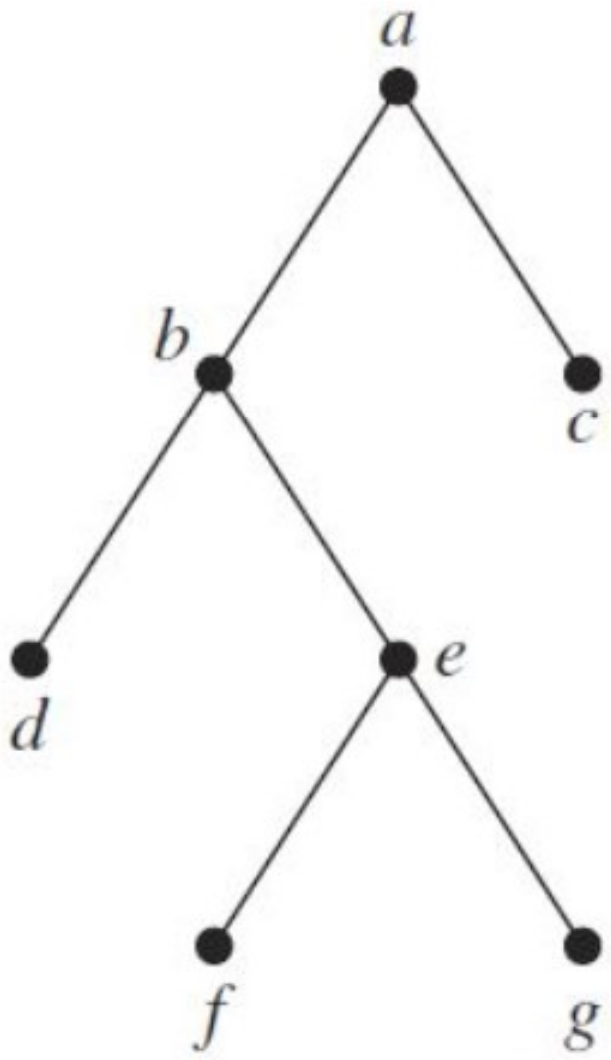








Exercise

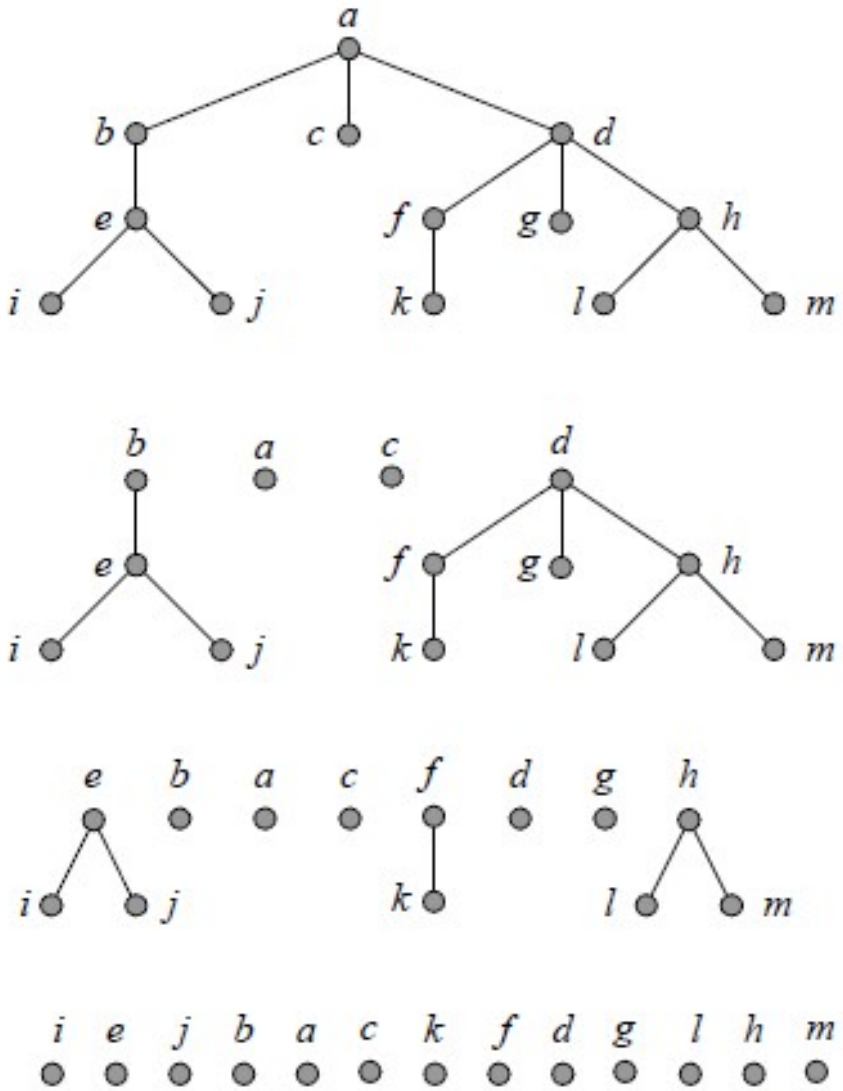


Determine the order in which a preorder traversal visits the vertices of the given ordered rooted tree.

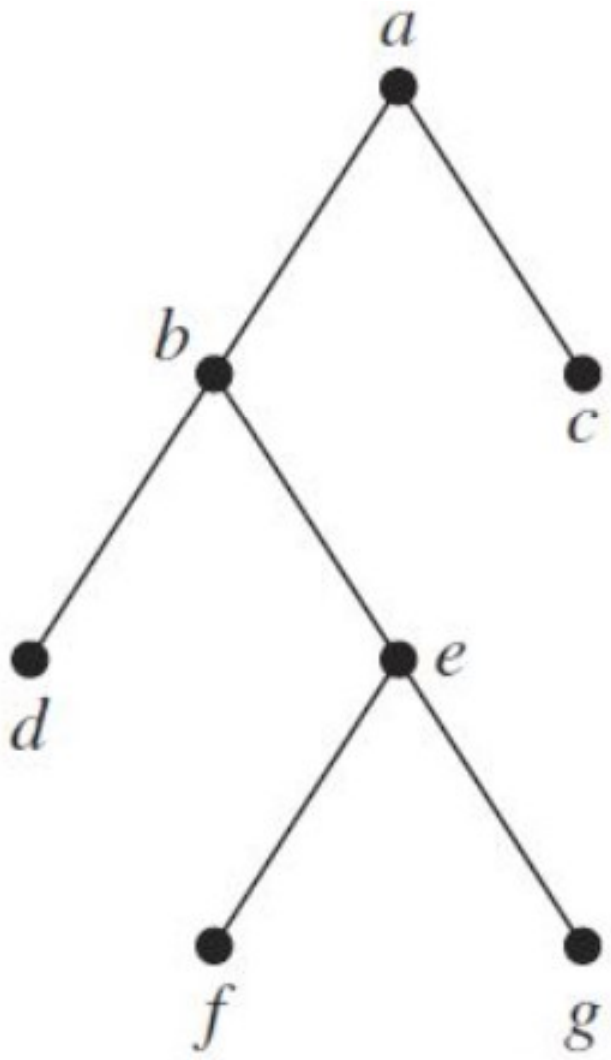
Inorder Traversal

```
Procedure inorder(T: ordered rooted tree)
r := root of T
If r is a leaf then list r
else
begin
    l := first child of r from left to right
    T(l) := subtree with l as its root
    inorder(T(l))
    list r
    for each child c of r except for l from left to right
        T(c) := subtree with c as its root
        inorder(T(c))
end
```

Inorder Traversal -Example



Exercise

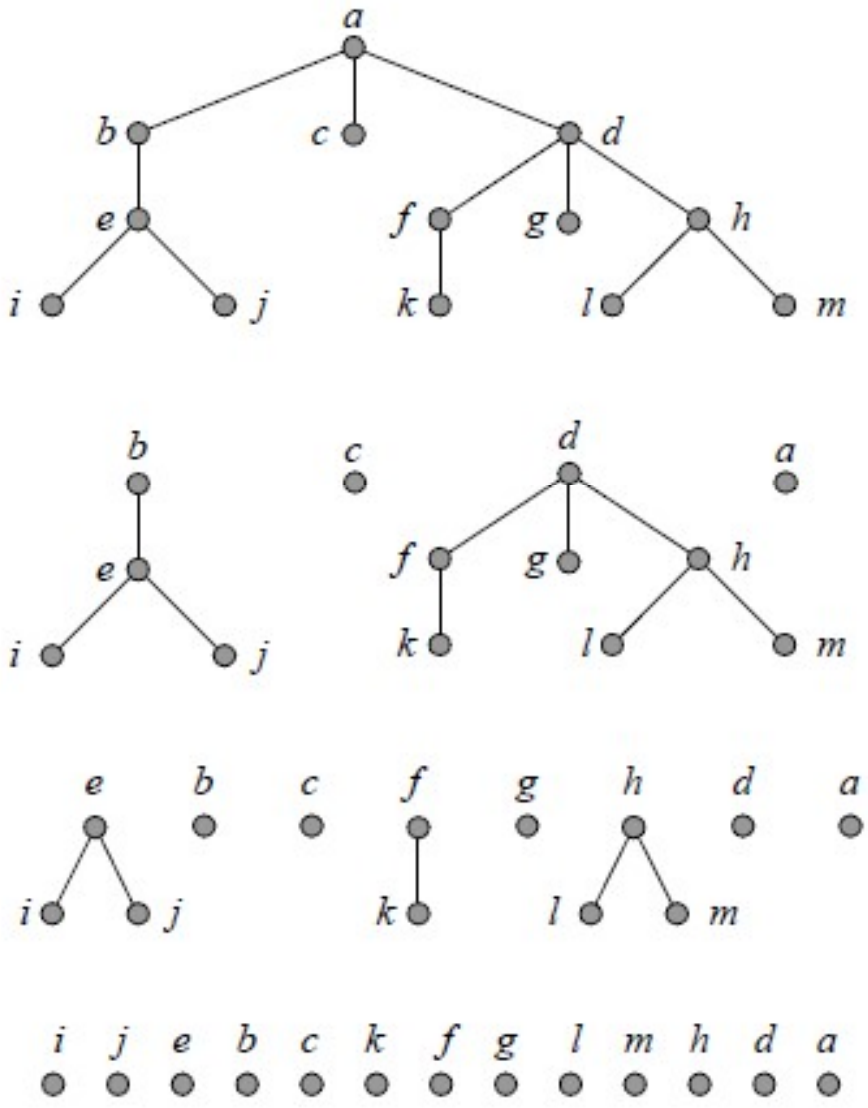


Determine the order in which a inorder traversal visits the vertices of the given ordered rooted tree.

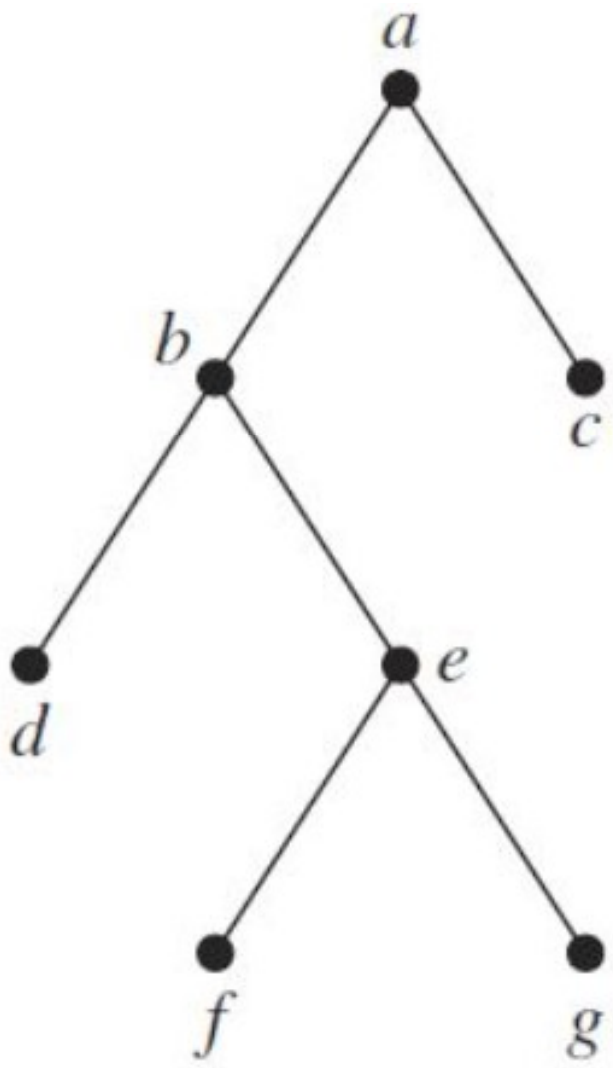
Postorder Traversal

Procedure *postorder*(*T*: ordered rooted tree)
r := root of *T*
for each child *c* of *r* from left to right
begin
    T(*c*) := subtree with *c* as its root
    postorder(*T*(*c*))
end
list *r*

Postorder Traversal - Example

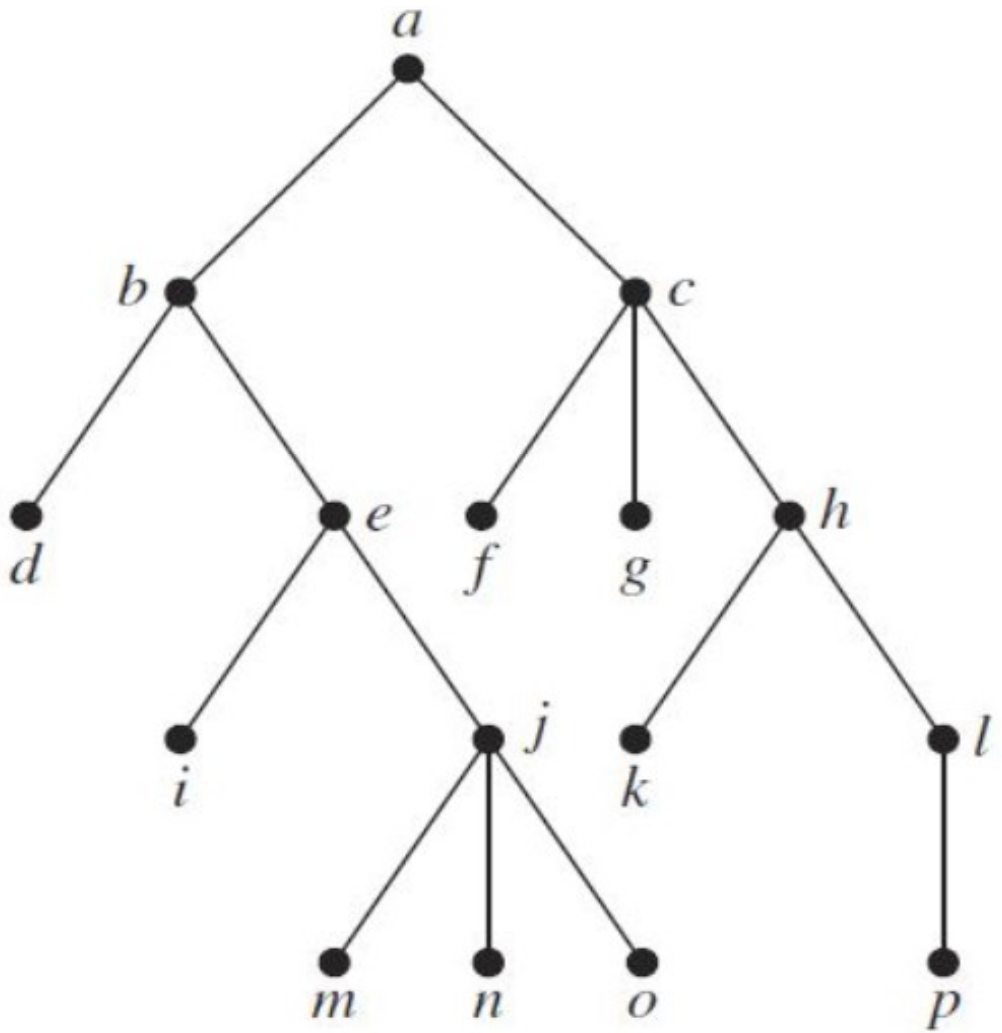


Exercise



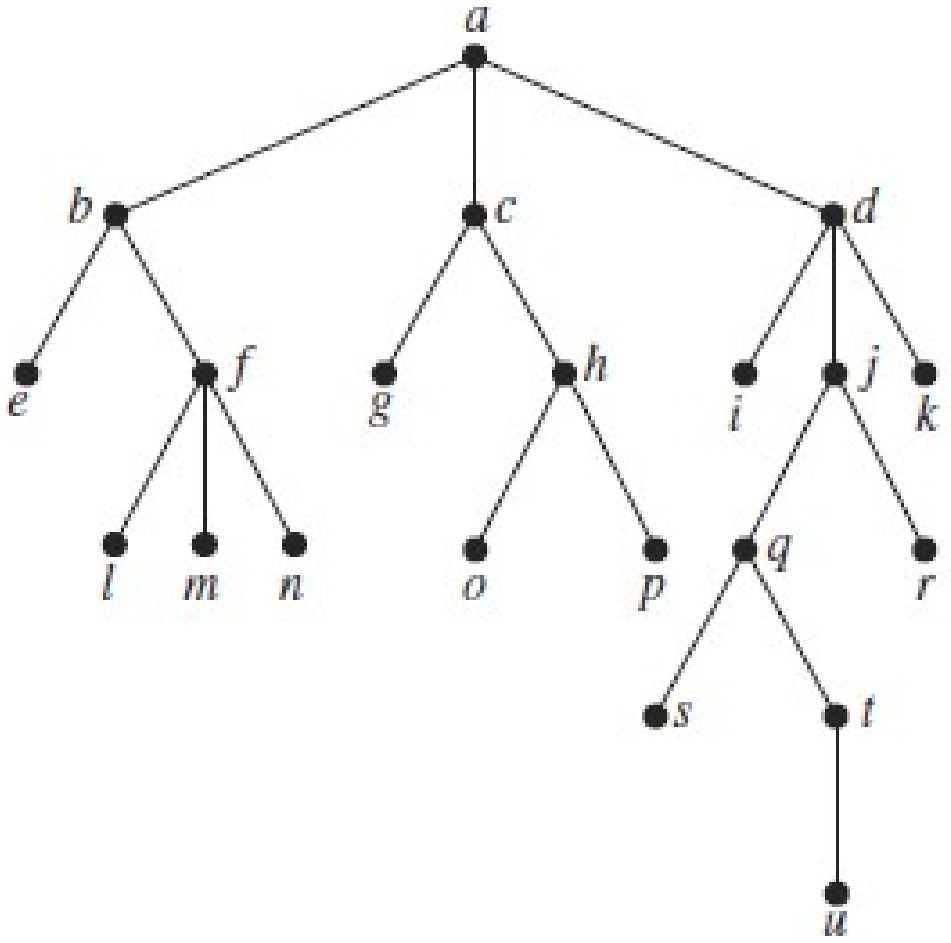
Determine the order in which a postorder traversal visits the vertices of the given ordered rooted tree.

Exercise



Determine the order of preorder, inorder and postorder of the given rooted tree.

Exercise



Determine the order of preorder, inorder and postorder of the given rooted tree.